MATH329 · OPERATIONS RESEARCH

# Midterm 462 Solutions

09/03/2025 · 30 marks

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## 01 Solution Q1: theory

### a The special cases of the graphical method

1. Alternative optimal solutions
2. Unbounded solution
3. No feasible solution

### b The two requirements to start simplex

**1. The coefficients of the basic variables (slack variables) form an Identity matrix.**

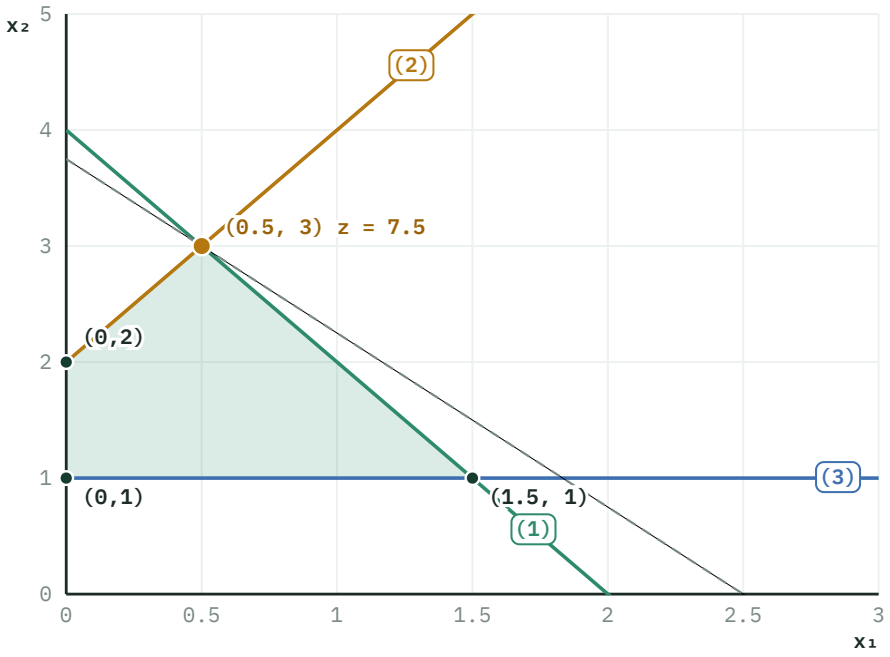
**2. In the objective function row, the coefficients of the basic variables are zero.**

## 02 Solution Q2: graphical

### 1 Prepare the constraints

| | | |
|---|---|---|
| (1) | $2x_1 + x_2 = 4$ | points (0, 4) and (2, 0) |
| (2) | $-2x_1 + x_2 = 2$ | points (0, 2) and (−1, 0) |
| (3) | $x_2 = 1$ | horizontal line |

Constraints (1) and (2) hold at $(0, 0)$, so their region is on the origin's side.
Constraint (3) is $\geq$, so its region is above the line $x_2 = 1$.




## 2 The corner points

| Corner | Where it comes from | $z = 3x_1 + 2x_2$ |
|---|---|---|
| (0, 1) | (3) with the $x_2$ axis | 2 |
| (1.5, 1) | (1) with (3) | 6.5 |
| **(0.5, 3)** | (1) with (2) | **7.5** |
| (0, 2) | (2) with the $x_2$ axis | 4 |

(1) + (2): $2x_2 = 6 \rightarrow x_2 = 3$ $\qquad$ $2x_1 + 3 = 4 \rightarrow x_1 = 0.5$

**$Z = 7.5$ at $x_1 = 0.5$ and $x_2 = 3$**

# 03 Solution Q3: standard form

## 1 The methods — 2 M

Graphical method and Algebraic method.

## 2 Standard form — 6 M

$4x_2 = -7 \quad \times (-1) \quad \rightarrow \quad -4x_2 = 7$

$x_2 = x_2^+ - x_2^-$ in every place $x_2$ appears

$$\text{Maximum } Z = 8x_1 - 2x_2^+ + 2x_2^- + 0s_1 + 0s_2$$

$$\text{S.to} \quad 2x_1 + x_2^+ - x_2^- + s_1 = 12$$

$$x_1 - 5x_2^+ + 5x_2^- - s_2 = 6$$

$$-4x_2^+ + 4x_2^- = 7$$

$$x_1, x_2^+, x_2^-, s_1, s_2 \geq 0$$

Marking: two marks for the objective function and one for each of the four lines.

# 04 Solution Q4: simplex for a min problem

## a Standard form 2 M

| | |
|---|---|
| Minimize $Z = x_1 - 2x_2 + 0s_1 + 0s_2$ | |
| S.to | $2x_1 + 3x_3 + s_1 = 1$ |
| | $3x_1 + 2x_2 - x_3 + s_2 = 5$ |
| | $x_1, x_2, x_3, s_1, s_2 \geq 0$ |

The key writes the last line as $x_1, x_2, s_1, s_2, s_3 \geq 0$, which is a typo: the problem has $x_3$ and no $s_3$.

## b Simplex 8 M

The $Z$ row comes from $z - x_1 + 2x_2 = 0$. The problem is min, so the entering variable is the most positive entry.

| Basic | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | Sol. | Ratio |
|---|---|---|---|---|---|---|---|
| $Z$ | $-1$ | $2$ | $0$ | $0$ | $0$ | $0$ | |
| $s_1$ | $2$ | $0$ | $3$ | $1$ | $0$ | $1$ | – |
| $s_2$ | $3$ | **$2$** | $-1$ | $0$ | $1$ | $5$ | $\frac{5}{2}$ |
| $Z$ | $-4$ | $0$ | $1$ | $0$ | $-1$ | $-5$ | |
| $s_1$ | $2$ | $0$ | **$3$** | $1$ | $0$ | $1$ | $\frac{1}{3}$ |
| $x_2$ | $\frac{3}{2}$ | $1$ | $-\frac{1}{2}$ | $0$ | $\frac{1}{2}$ | $\frac{5}{2}$ | – |
| $Z$ | $-\frac{14}{3}$ | $0$ | $0$ | $-\frac{1}{3}$ | $-1$ | $-\frac{16}{3}$ | |
| $x_3$ | $\frac{2}{3}$ | $0$ | $1$ | $\frac{1}{3}$ | $0$ | $\frac{1}{3}$ | |
| $x_2$ | $\frac{11}{6}$ | $1$ | $0$ | $\frac{1}{6}$ | $\frac{1}{2}$ | $\frac{8}{3}$ | |

Iteration 1: the most positive entry is $2$ under $x_2$. The $s_1$ row has a zero coefficient, so it has no ratio, and the $s_2$ row has ratio $\frac{5}{2}$. So $s_2$ leaves and the pivot is $2$.

Iteration 2: a positive $1$ remains under $x_3$. The $s_1$ row has ratio $\frac{1}{3}$, and the $x_2$ row has a negative coefficient. So $s_1$ leaves and the pivot is $3$.

**Optimum $z = -\frac{16}{3} = -5.33$ at $x_1 = 0$, $x_2 = \frac{8}{3} = 2.67$, $x_3 = \frac{1}{3} = 0.33$**

Check: $z = 0 - 2(\frac{8}{3}) = -\frac{16}{3}$

## The same tableaus in decimals (as in the answer key)

| Basic | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | Sol. |
|---|---|---|---|---|---|---|
| Z | −1 | 2 | 0 | 0 | 0 | 0 |
| $s_1$ | 2 | 0 | 3 | 1 | 0 | 1 |
| $s_2$ | 3 | **2** | −1 | 0 | 1 | 5 |
| Z | −4 | 0 | 1 | 0 | −1 | −5 |
| $s_1$ | 2 | 0 | **3** | 1 | 0 | 1 |
| $x_2$ | 1.5 | 1 | −0.5 | 0 | 0.5 | 2.5 |
| Z | −4.67 | 0 | 0 | −0.33 | −1 | −5.33 |
| $x_3$ | 0.67 | 0 | 1 | 0.33 | 0 | 0.33 |
| $x_2$ | 1.83 | 1 | 0 | 0.17 | 0.5 | 2.67 |

First table 2m (every row 0.5m + 0.5m for pivot element).

Second and third table 3m (1m for every row). Every mistake in calculation −0.5m.